Evaluation Parameters

Image Quality:

PSNR - Peak Signal-to-Noise Ratio
SSIM - Structural Similarity Index Measure
FID - Fréchet Inception Distance

Qualitative:

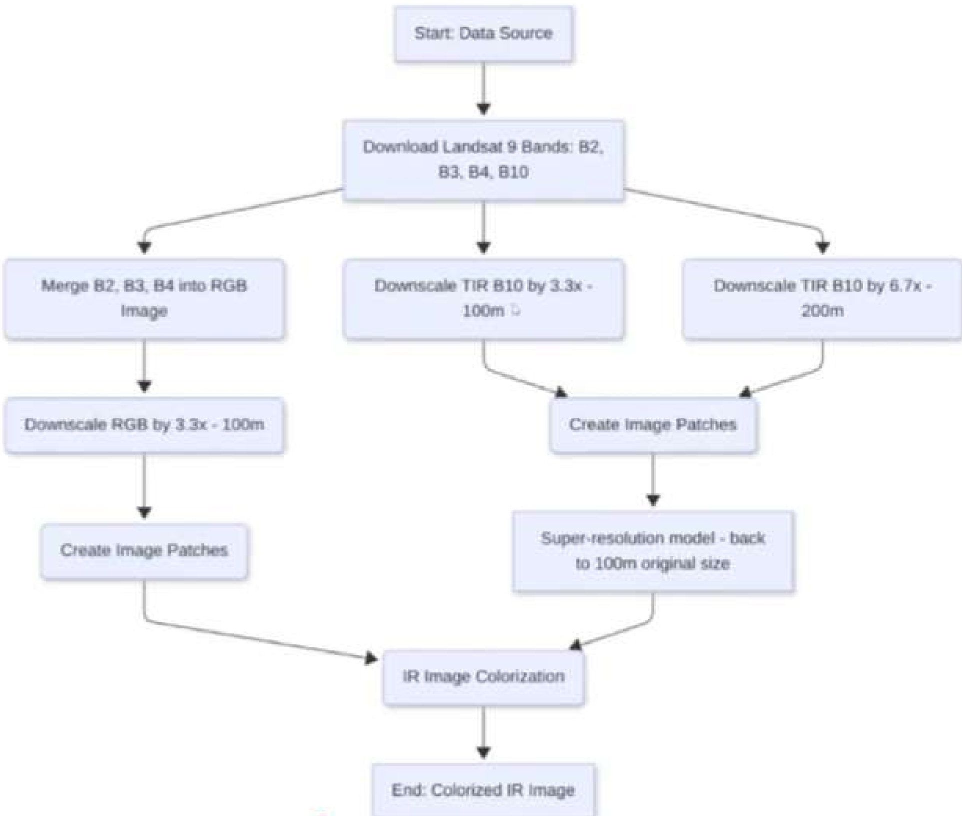
Visual inspection to prevent 'hallucinations'

Preferred:

Low inference time per tile.

Bonus: Can you design and implement **physics informed modeling** for the task?

WorkFlow Summary



Expected Output

An end-to-end trained Deep Learning pipeline.

Input:

Single-channel TIR image @200m.

Output:

1) High-resolution TIR image @ 100m.
2) colorized RGB image @ 100m.

Dataset & Tech stack

Dataset: Landsat 9 (USGS) Processed TIR1-RGB pairs using our github instructions.

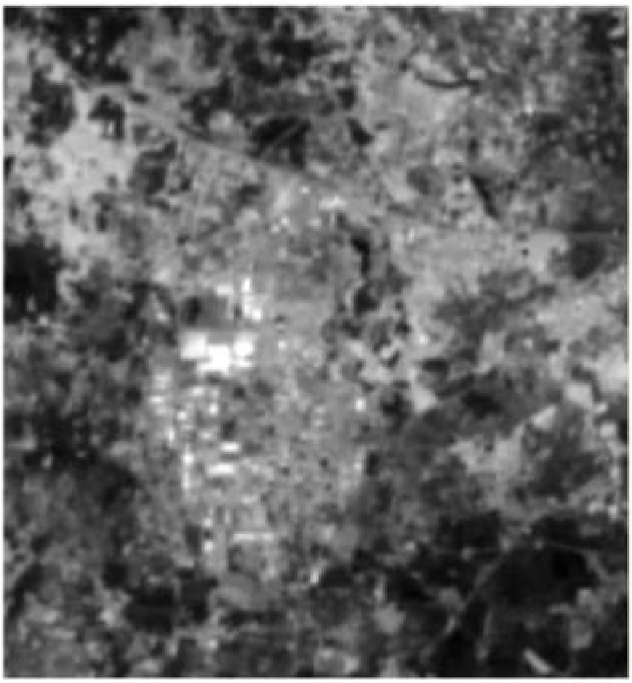
Framework: Python-based end-to-end solution.

Models: GANs/Diffusion Models/SoTA Image-to-Image models.

Libraries: GDAL, Rasterio, tifffile, OpenCV.

Landsat 9 Imagery @ 30m Resolution

RGB



TIR-1 (actual sensor resolution is 100m)

Problem Statement

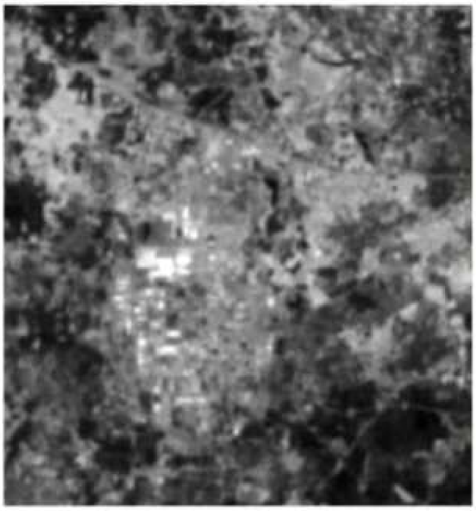
Develop an end-to-end framework that super-resolves structural details and predicts realistic colorization (RGB translation) for Thermal Infrared satellite Images.

Objective

Super-Resolution: Objectively improve the resolution of Thermal Infrared (TIR) inputs.

Colorization: Map TIR features to realistic RGB representations using generative AI methods.

Preserve structural and spectral Integrity!



Sample Landsat-9 TIR-1 Image

- TIR images are vital for Earth observation but have low resolution compared to visible bands.
- Difficult for humans and AI to interpret using traditional pipelines.